



Databases 35m

What is a database?

A database is software that accepts and stores data in a predefined structure or arrangement. Databases can vary in complexity, ranging from simple storage on your computer or smartphone to massive data centres for the likes of Facebook and Google. This topic will emphasise using a database specifically for the benefit of a web application. The database is responsible for storing and supplying data in a structured manner.

Some key aspects of a database are:

- Schema – describes how the data is to be stored
- Views – a specific view of the data based on a stored query
- Tables – the part where the actual data is stored, often arranged in columns and rows (more on this later)
- Queries – the language specific to a particular database; often in structured query language (SQL).

A database would, in most cases, require an additional piece of software, such as a database management system (DBMS), to interact with and manage the database.

In a web application, a database will store and display updated information. Examples of web applications that depend on databases are registration websites, discussion forums and e-commerce websites.

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Types of databases

We've previously introduced you to the two types of databases that you will encounter as a web developer: a relational database management system (or SQL database) and a non-relational database management system (or NoSQL database). We will now go into detail about these databases.



If you need a refresher about this, we recommend taking a look at the following page:

Database technologies →

Relational database (RDB)

This is a database based on the relational model (Codd, 1970)^{fnref:15184b} where data is arranged using a specified structure and language. All data is arranged as tuples, or data records in rows, and grouped based on their relationships to form a table.

Each tuple is headed with attributes, or what is typically referred to as field names. The RDB is one of the most common database types. MySQL, the one used in this course, is the most popular open-source database available (DB-Engines, n.d.)^{fnref:191d92} and is used by a good number of web applications on the internet. MySQL is used by companies such as Netflix, Spotify and YouTube to collect and store their data.

NoSQL

NoSQL means 'not only SQL', which means that they may use SQL, but they also use other access mechanisms. Their architecture is different from relational databases because they store data as documents, which are structured using formats like JavaScript Object Notation (JSON). This allows them to be much more flexible than relational databases in terms of how they store data because they do not rely on having a fixed schema. This lack of rigidity results in faster performance and easier replication across data centres. Their flexibility and speed make them ideal for contemporary systems that need to constantly scale and adapt.

There are several different types of NoSQL databases, including:

➤ **Skip to main content** ➤ **Document databases that store JSON, XML or other types of documents**

- key-value stores that use unique keys, like relational databases, to retrieve data from collections
 - wide-column databases that use a table format but are very flexible about the data that can be stored in the table columns
 - graph databases that can identify patterns in unstructured and semi-structured information.
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Popular NoSQL databases include Redis, MongoDB, Apache Cassandra, RavenDB (for .NET), DynamoDB (for Amazon Web Services), HyperTable (based on Google's Bigtable design) and Firebase Realtime Database (part of Google's Firebase development platform).

Read *Overview of Modern NoSQL Database Management Systems. Case study: Apache Cassandra* (Ferencz, 2018), [fnref:09632a](#) to understand more about NoSQL databases and when it is more feasible to use NoSQL databases over traditional SQL-based DBMSs.

Remember the following videos are optional.

Dig deeper

Watch the playlist MySQL tutorial for beginners, created by Bro Code (2022). [fnref:3728eb](#) It consists of **31 videos** that cover essential topics such as:

- **Introduction and installation:** Setting up MySQL on your system
- **Database and table creation:** Step-by-step guides to creating databases and tables
- **Data manipulation:** Techniques for inserting, selecting, updating and deleting data
- **Constraints and keys:** Understanding constraints like UNIQUE, NOT NULL, CHECK and DEFAULT and keys such as PRIMARY and FOREIGN
- **Advanced features:** Exploring joins, functions, logical operators, wildcards and

Skip to main content like ORDER BY and LIMIT

- **Complex queries:** Learning about subqueries, groupings, rollups and unions
- **Stored procedures and triggers:** Implementing stored procedures and triggers for advanced database management.

This playlist is ideal for anyone looking to build a solid foundation in MySQL.



☒ SQL versus NoSQL databases

Non-Assessed

Describe the problems with SQL databases that NoSQL databases solve.

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Let's now investigate typical database features.



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Thoughts on this page?



Databases

Have you used any of the databases mentioned in our discussion? We'd love to hear about your experiences! Do you have a favourite between SQL/NoSQL?

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